

FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES

DEPARTMENT OF ECONOMICS

ECONOMICS 114 A2

DATE:	25 May 2023
DURATION:	3 hours
TOTAL OF MARKS:	100
EXAMINERS:	Prof S du Plessis (convenor) Ms L Boltman Dr C Links Ms W Nkosi Dr D Shepherd Mr M van Schoor
INTERNAL MODERATOR:	Prof P de Villiers

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER **SECTION A**'s QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER **SECTION B**'s QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS NEXT** TO THE NUMBER OF THE STATEMENT.

ANSWER **SECTION C**'s QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS

KEY:

AP = Average product
AC = Average cost
BL = Budget line
D = Demand
DWL = Deadweight loss
FF = Feasible frontier
GDP = Gross domestic product
IC = Indifference curve
K = Capital
L = Labour
MB = Marginal benefit
MC = Marginal cost
MEB = Marginal external benefit
MP = Marginal product
MPB = Marginal private benefit
MPC = Marginal private cost

MR = Marginal revenue
MRS = Marginal rate of substitution
MRT = Marginal rate of transformation
MSB = Marginal social benefit
MSC = Marginal social cost
P = Price
PF = Production frontier
Q = Quantity
Q_d = Quantity demanded
Q_s = Quantity supplied
S = Supply
TC = Total cost
TP = Total product
TR = Total revenue
w = Wage
Y = Income

SECTION A
TRUE OR FALSE STATEMENTS

[10 marks: 1 mark for a correct answer and - ½ for an incorrect answer]

1. It is impossible for a game to have two Nash equilibria.
a) True
b) False
2. Players in a repeated game may be able to solve social dilemmas.
a) True
b) False
3. In a voluntary exchange between two parties, the Pareto efficiency curve shows allocations that make participants better off than they would be without an exchange.
a) True
b) False
4. It can be considered procedurally fair that people who work hard earn higher incomes than those who do not work as hard.
a) True
b) False
5. Employers generally write all tasks that an employee will be required to perform in an employment contract.
a) True
b) False
6. In the labour discipline model, a firm's profit level will be the same if employee effort per hour and the hourly wage both increased by 15%.
a) True
b) False
7. If a firm is currently experiencing economies of scale, its average cost of production decreases as production is increased, assuming all inputs are variable.
a) True
b) False
8. A competitive firm is currently earning zero economic profit. Suppose there is then a decrease in market demand. This firm is likely to increase the quantity of output it produces.
a) True
b) False

9. Marginal social cost is the cost of producing an additional unit of a good that is incurred by anyone other than the producer of the good.
- True
 - False
10. Pigouvian taxes shift the external costs borne by society to the producers of these externalities.
- True
 - False

SECTION B

MULTIPLE CHOICE QUESTIONS

[30 marks: 2 marks for a correct answer and – ¼ for an incorrect answer]

11. RESEP and LEAP are two research units in the Department of Economics at Stellenbosch University (SU). In the wake of SU's phased return to campus after the Covid-19-related lockdown restrictions were eased, both RESEP and LEAP must decide whether they want to return to campus, or whether they want to continue working from home. The decision by each research unit is made independently, and without consulting the other team. The interaction is represented below.

		LEAP	
		Return to campus	Work from home
RESEP	Return to campus	8, 8	14, 4
	Work from home	4, 14	12, 10

Which of the following statements regarding the game above are **CORRECT**?

- (Return to campus, Return to campus) is a Nash equilibrium in this game.
 - (Work from home, Work from home) is a dominant strategy equilibrium in this game.
 - This game is an example of an invisible hand game.
 - The socially optimal outcome is not the predicted equilibrium of this game.
- i and iv
 - ii and iv
 - i and iii
 - i, ii and iii

12. Detectives Jake Peralta and Rosa Diaz are interviewing a suspect they have arrested for a string of robberies in Brooklyn and need to get a confession out of their suspect. Jake and Rosa can either choose to be **empathetic** or **tough**. Unfortunately, because the police station is very busy, Jake and Rosa cannot discuss their approach with one another, so they will have to make this decision independently. The interaction is represented by the table below.

		Rosa	
		Empathetic	Tough
Jake	Empathetic	3, 2	3, 3
	Tough	4, 2	2, 5

Which of the following statements regarding the game above are **CORRECT**?

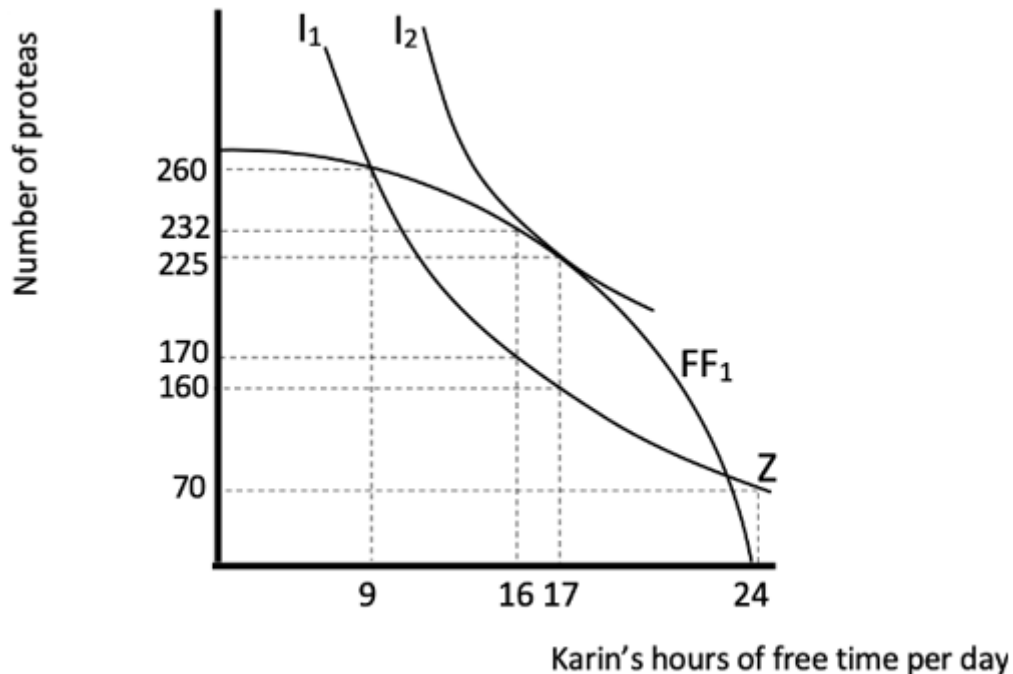
- The Nash equilibrium of the game is (Tough, Empathetic).
 - Jake has a dominant strategy to be Empathetic.
 - This game represents a prisoner's dilemma.
 - Rosa has a dominant strategy to be Tough.
- i and iii
 - iii and iv
 - iv only
 - ii, iii and iv
13. Troy and Abed need to bring one type of snack to a party. They both favour pizza, but they generally prefer to have a variety of party food rather than eating only pizza. The payoff matrix below represents their strategic interaction:

		Abed	
		Pizza	Doughnuts
Troy	Pizza	2, 2	4, 5
	Popcorn	5, 4	3, 3

Which one of the following statements is **INCORRECT**?

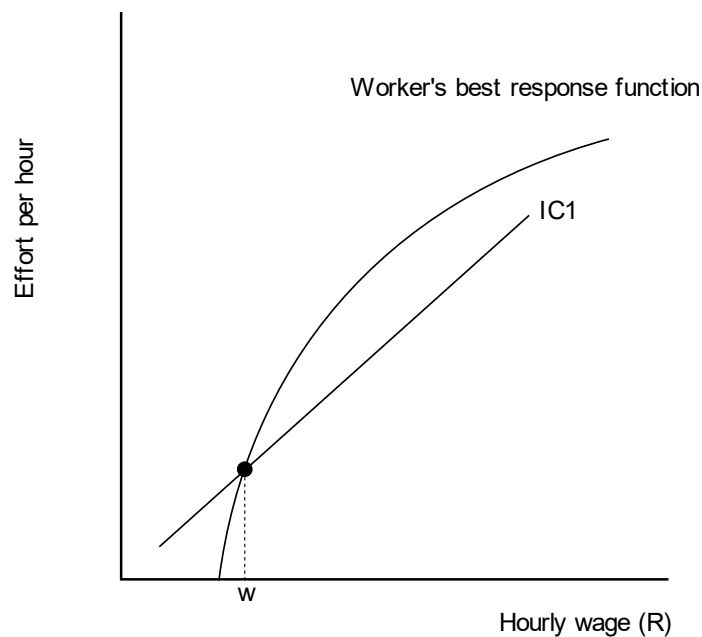
- (Pizza, Pizza) is Pareto dominated by (Popcorn, Doughnuts).
- There are three Pareto-efficient outcomes in this game.
- (Popcorn, Pizza) is a Pareto-efficient outcome.
- Neither Troy nor Abed has a dominant strategy.

14. Karin grows king proteas on Jessica's land but must pay Jessica a portion of her crop for the right to do so. The maximum possible production is represented by the combined feasible frontier (FF_1) and Karin's reservation indifference curve (I_1). Karin's preferences are quasilinear. Which one of the following statements is **CORRECT**?



- a) Jessica will maximise her surplus when she asks for a rent of 62 proteas and Karin chooses to work 7 hours a day.
- b) Karin would never choose to work more than 15 hours a day.
- c) It would be Pareto efficient for Karin to work 7 hours a day and to pay Jessica 50 proteas.
- d) The number of proteas that must be sacrificed exceeds the number of proteas that Karin is willing to sacrifice when she increases her hours of free time from 16 to 17 hours.

15. A worker is currently being paid an hourly wage of w . The graph below indicates her best-response curve and an isocost curve for her employer (IC_1):



Which one of the following statements is **CORRECT**?

- a) A small increase in the hourly wage will result in an increase in the effort-to-wage ratio.
 - b) A small decrease in the hourly wage will result in an increase in the employee's employment rent.
 - c) A small increase in the hourly wage will result in a decrease in the firm's profit.
 - d) The firm can increase profit by as much as it wants to by increasing wages.
16. Labour contracts are different to contracts for goods and services because:
- a) labour quality can be better monitored.
 - b) labour contracts often have a longer duration.
 - c) work tends to be more standardised.
 - d) workers tend to be indifferent between keeping and losing their jobs.
17. Sylvia is a private security guard, currently earning R80 per hour and working 10 hours per day. She spends R40 per day on bus tickets to commute to and from her job. Her uncle owns a bakery where she can always work for a wage of R50 per hour (also for 10 hours per day) with no transport costs. What is her current employment rent?
- a) Zero
 - b) R260 per day
 - c) R800 per day
 - d) Minus R500 per day

18. Suppose a firm has the following inverse demand function, total revenue function and total cost function:

$$\begin{aligned}P &= 100 - 2Q \\TR &= 100Q - 2Q^2 \\TC &= 20Q + 200\end{aligned}$$

Which one of the following is **INCORRECT** for the given demand, total revenue, and total cost functions?

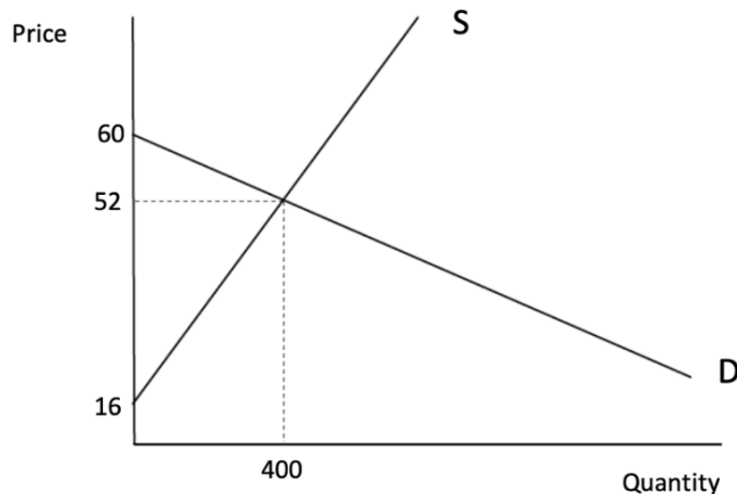
- a) The marginal revenue function is $MR = 100 - 4Q$.
 - b) The average cost function is $AC = 20 + 200/Q$.
 - c) The marginal cost function is $MC = 20 + 200$.
 - d) The profit-maximising quantity of output is 20 units.
19. The demand and supply of tobacco per kilogram are given by the following functions:

$$\begin{aligned}Q_d &= 100 - 2P \\Q_s &= 3P - 30\end{aligned}$$

What will happen in the tobacco market if the government imposes a price ceiling of R20 per kilogram of tobacco?

- a) There will be a shortage of 30 kilograms.
 - b) There will be a surplus of 30 kilograms.
 - c) There will be a shortage of 60 kilograms.
 - d) The equilibrium price and quantity will remain the same.
20. You are a vendor selling doughnuts on the Rooiplein. One day, you decide to raise the price of your doughnuts from R2 to R2.80 each. As a result, the quantity demanded of your doughnuts falls from 100 to 80 per day. Assume that the demand for your doughnuts is represented by a linear function. This decision results in a _____ in your total revenue because your elasticity of demand is _____.
- a) increase; 1.2
 - b) increase; 0.5
 - c) decrease; 0.5
 - d) decrease; 1.2

21. The market for coffee has many sellers and buyers, all of whom are price takers. A major storm has hit the coffee-growing regions, causing significant damage to the coffee crops. Which one of the following statements is **CORRECT**?
- The market supply curve will shift to the right and the market demand curve will shift to the left. As a result, the equilibrium quantity will increase.
 - The market supply curve will shift to the left, the equilibrium price of coffee will increase, and the equilibrium quantity will decrease.
 - The market supply curve will shift to the left and the market demand curve will shift to the right. As a result, the equilibrium price of coffee will increase.
 - The demand curve will shift to the left and the equilibrium quantity will increase.
22. A firm operates in a perfectly competitive market. Which one of the following statements is **INCORRECT**?
- There is no profit incentive for this firm to charge a price higher than the equilibrium price.
 - This market is efficient, but the outcome is not necessarily fair.
 - The firm will produce where its negatively sloped demand curve intersects the positively sloped MC curve.
 - Total surplus is maximised in this market.
23. The graph below shows the demand and supply curves of a certain good.



The government introduces a per unit tax of R22 on suppliers. The price paid by consumers following the introduction of this tax is

- R56
- R58
- R60
- R74

24. In Papua New Guinea the disposal of mine tailings polluted the rivers and had a devastating effect on cocoa production. Which one of the following statements regarding marginal costs is **CORRECT**?
- a) Marginal private cost (MPC) is the marginal cost imposed on cocoa producers by the mines.
 - b) Marginal social cost (MSC) is the marginal cost of the polluted rivers for cocoa producers.
 - c) Marginal social cost (MSC) is the difference between the MPC and the Marginal external cost (MEC).
 - d) Marginal social cost (MSC) is the marginal cost to society of mining operations.
25. Which one of the following is a negative externality connected to attending university?
- a) A university degree acts as a signaling mechanism to employers.
 - b) Other costs, such as books and materials, that are incurred in addition to tuition and fees.
 - c) The university requires that all individuals living in student housing must either show that they have already received vaccinations against all communicable diseases, or they must receive these vaccinations before they move into student housing.
 - d) The people in the room next door that play loud music at times when you want to sleep.

SECTION C
DESCRIPTIVE QUESTIONS
[60 marks]

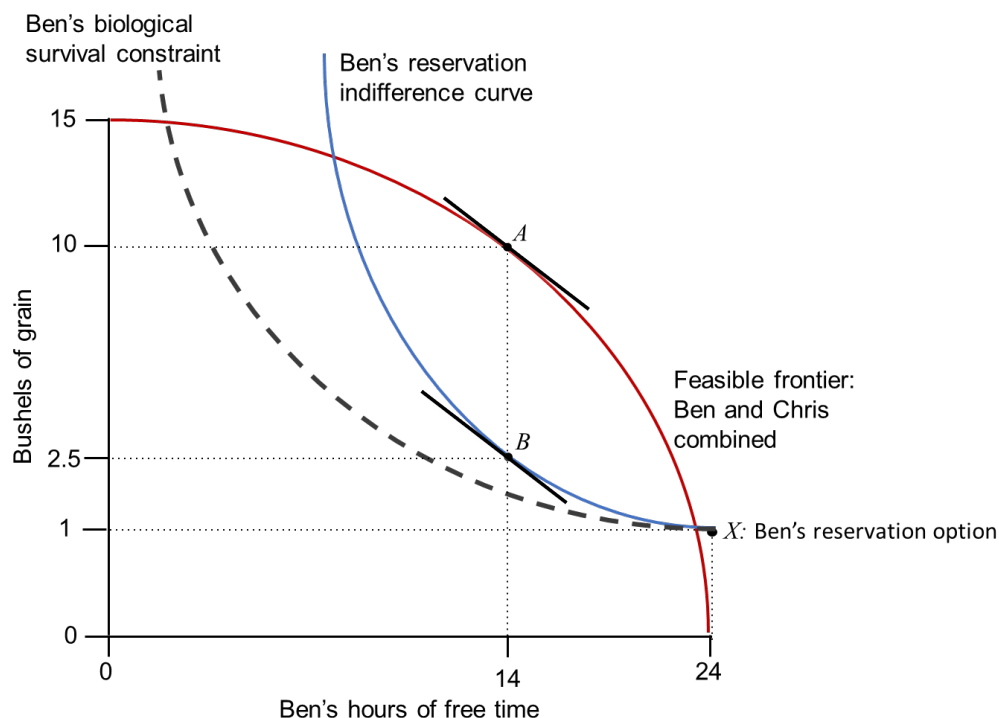
QUESTION 26 [4 marks]

Consider the following ultimatum game between two players, Sindisiwe and Jane. Sindisiwe is given R1 000 and must offer a portion of it to Jane. Jane can either accept or reject the offer. If Jane accepts the offer, both players get to keep their share of the money as offered by Sindisiwe. If Jane rejects the offer, neither player gets any money.

- 26.1 Suppose Sindisiwe offers R300 to Jane. What is Jane's best response if all she cares about is the money received in this game? Explain briefly. [2]
- 26.2 What factors in the real world might influence a Responder to accept or reject an offer from a Proposer? [2]

QUESTION 27 [4 marks]

Chris is a landowner and Ben is a farmer who pays a share of his grain output to Chris for the use of the land. The figure below shows Chris and Ben's feasible frontier, Ben's biological survival constraint and his reservation indifference curve. Ben has quasi-linear preferences.



27.1 Is the following statement **TRUE** or **FALSE**? Explain your answer.

“For any feasible number of hours of free time, the marginal rate of substitution on the reservation indifference curve is larger than that on the biological survival constraint.” [2]

27.2 How many hours does Ben have to work to maximise total economic rent and what is the maximum total economic rent? [2]

QUESTION 28 [5 marks]

Ntsako is employed as a journalist at CapeNews, an Internet-based news publication. He earns R100 per hour and works reasonably hard, but he still spends part of his time at work reading novels rather than working.

28.1 Explain why CapeNews could obtain more effort from Ntsako if they increased his wage. Your answer should refer to the impact on his employment rent. [4]

28.2 Suppose after some time in his job, Ntsako realises that he loves working with his colleagues on interesting news stories. What is the effect of this on his reservation wage? [1]

QUESTION 29 [5 marks]

You are working as an economist for Milk South Africa. You are asked to assess the impact of two events (that occur simultaneously) on the market for milk:

- A severe drought causes dairy farmers to reduce the number of milk-producing cattle by a third.
- A new report by the South African Medical Association reveals that the consumption of milk has significant health benefits.

With the aid of a graph, explain how these events impact the equilibrium price and quantity of milk. [5]

QUESTION 30 [11 marks]

Frank's Furters is a producer of gourmet hotdogs in Cape Town. Although there are many other firms producing hotdogs, *Frank's Furters* product is different from its competitors. The firm is therefore able to set its own price in the market for hotdogs. The demand for *Frank's Furters* hotdogs is given by:

$$Q_d = 340 - 4P$$

The firm's marginal and total cost functions are given by:

$$MC = \frac{1}{8}Q + 5$$

$$TC = 250 + \frac{1}{16}Q^2 + 5Q$$

- 30.1 What is *Frank's Furters*' fixed cost of producing hotdogs? [1]
- 30.2 What is *Frank's Furters*' profit-maximising price and quantity? Show all your calculations. [5]
- 30.3 Is *Frank's Furters* making a positive or negative economic profit? Show all your calculations. [3]
- 30.4 Do you expect there to be more or fewer firms selling hotdogs in the future? Briefly explain. [2]

QUESTION 31 [7 marks]

A bookstore wants to increase its total revenue. One strategy is to offer a 10% discount on every item it sells. The bookseller sells two products, books and coffee. The table below shows how consumers of the two products respond to the discount.

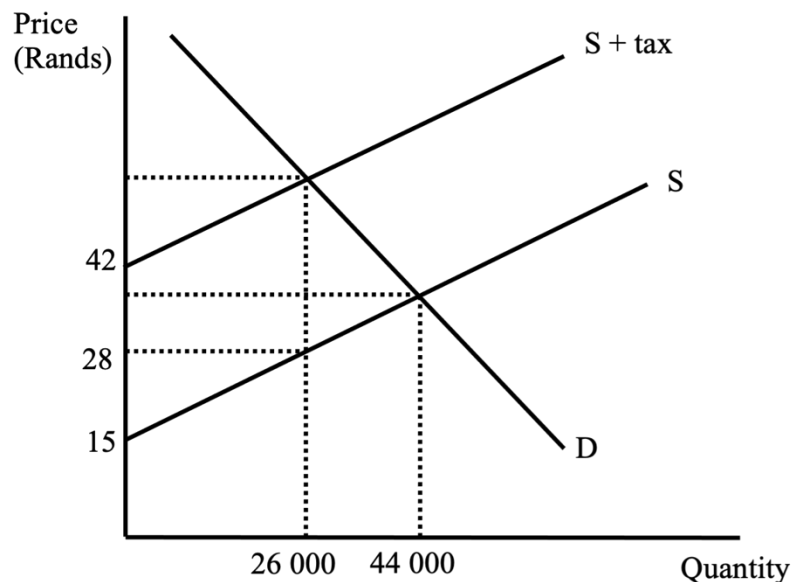
	Books	Coffee
Quantity sold before the 10% discount	1.55 million	1.50 million
Quantity sold after the 10% discount	1.65 million	1.70 million

- 31.1 Calculate the price elasticities of demand for the two products. [4]

- 31.2 Would you advise that the bookstore offers a discount on both products? Explain by referring to your answers in 31.1. [3]

QUESTION 32 [7 marks]

The figure below depicts the competitive market for pizza. The government decides to impose a per unit tax on pizza, leading to a reduction in monthly pizza sales from 120 000 to 65 000.



- 32.1 What is the size of the per unit tax on pizza? [1]
- 32.2 What is the price paid by buyers of pizza following the imposition of the tax? [1]
- 32.3 Calculate the size of the deadweight loss due to the tax. [2]
- 32.4 Who bears the larger tax burden: buyers or sellers? Why? No calculations are required. [2]
- 32.5 How much tax revenue is generated for government? [1]

QUESTION 33 [11 marks]

The equations below represent the market demand and supply of cooking oil:

$$Q_d = 325\,000 - 2\,500P$$

$$Q_s = 500P - 5\,000$$

where Q_d and Q_s represent two-litre bottles of cooking oil. The government imposes a price ceiling of R80 on bottles of cooking oil.

- 33.1 Is this price ceiling binding? Show how you arrived at your answer. [3]
- 33.2 Does the price ceiling lead to a surplus or shortage in the market for cooking oil, and what is the size thereof? [3]
- 33.3 How does the price ceiling affect the size of the producer surplus? No calculations are required. [1]
- 33.4 Would a government subsidy given to producers that reduces the costs of producing cooking oil, help to eliminate any deadweight losses resulting from this price ceiling? Explain your answer. [4]

QUESTION 34 [6 marks]

When peach canners in Ceres process fresh peaches, they produce three products. The first (canned peaches) is sold in the marketplace. The other two (liquid and solid wastes) are by-products that must be removed. The liquid is sometimes temporarily kept in holding ponds and later released into a nearby stream or sewer. Liquid dumped in the stream represents a negative externality to downstream users. In the peach growing region, the marginal external costs (mec) of the canning process have been estimated as:

$$MEC = 0.000043Q$$

where Q represents the output of canned peaches in tins per week. The marginal private cost of canning peaches (ignoring MEC) is:

$$MC = 2 + 0.000157Q$$

and the market demand for canned peaches is:

$$P = 9 - 0.000243Q$$

- 34.1 How many tins of peaches will be produced per week during the growing season, and what will the selling price per tin be if producers ignore the costs imposed on others? [2]
- 34.2 If producers are forced to incorporate the marginal external costs into their production decisions by, for example, levying a Pigouvian tax, what will the new quantity and selling price be? [3]
- 34.3 If producers are forced to incorporate the marginal external costs into their production decisions, what will the effect on market efficiency be? [1]